

IAM Validation Scorecard

Complete Test Ledger — All Levels, All Scripts
Informational Actualization Model (IAM)

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Physics Constraint (Read First)

Critical implementation note, confirmed March 5, 2026: $H_{\text{matter}}^2(a) = H_{\Lambda\text{CDM}}^2(a) + \beta_m \mathcal{E}(a) H_0^2$ is the *matter-sector perturbation friction term only*. The background Friedmann equation is always standard ΛCDM . The μ - Σ ratio is $\mu(a) = H_{\Lambda\text{CDM}}^2/H_{\text{matter}}^2 < 1$. Applying this term to the background yields $H_0 \approx 61.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (unphysical); this was confirmed by the Level 2b diagnostic and rules out background-level modification.

1 Key Results Summary

Quantity	Value
β_m predicted (virial)	$\Omega_m/2 = 0.1575$
β_m from MCMC (Level 2a)	0.157 ± 0.020
Agreement	0.3%
$H_0(\text{matter})$	$72.51 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (0.51σ from SH0ES)
σ_8 (IAM)	0.800 (suppressed from 0.809)
μ_0 predicted	-0.136
Σ	1.0 exactly ($\delta\varphi = 0$)
Level 1 $\Delta\chi^2$ (Planck)	+1.43 to +2.32 (all 4 combos)
Level 2a $\Delta\chi^2$ (Planck best-fit)	+0.54
Derivation suite $\Delta\chi^2$ (demo set)	+31.2 (5.6σ , 0 free params)
ΛCDM relative likelihood	$6 \times 10^6 \times$ less likely
Convergence (all chains)	$R - 1 < 0.01$
Euclid DR1 projected significance	$\sim 3.4\sigma$ on μ_0

2 Section 1: Level 1 — MGCAMB μ - Σ Validation

Tool: MGCAMB v1.5.2 + Cobaya, pure_MG parametrization. **Chains:** Runs A–L (12 chains). **Status:** **CONFIRMED**.

#	Test	Result	Key Diagnostic
1	χ^2 viability vs. ΛCDM	PASS	$\Delta\chi^2 = +1.43$ to $+2.32$ (all 4 combos)
2	β_m posterior	PASS	0.157 ± 0.029 (68%); matches virial to 0.3%
3	β_γ constraint	PASS	$\beta_\gamma < 1.40 \times 10^{-6}$ (95% CL)
4	Standard-param stability	PASS	All shifts $< 0.1\sigma$
5	σ_8 suppression	PASS	$0.8087 \rightarrow 0.7998$ (real growth, not rebalancing)
6	Posterior shape	PASS	Gaussian, no multimodality
7	7/7 Boltzmann diagnostics	PASS	All solver consistency checks pass
8	$R - 1$ convergence	PASS	< 0.01 all chains

3 Section 2: Level 2a — Dual-Sector CAMB Validation

Tool: Vanilla CAMB v1.5.8, modified Fortran (`equations_iam_level2.f90`, 3 surgical changes to `adotoa_matter`). **Chains:** Runs A, C, D (3 chains). **Status:** **CONFIRMED**.

Modification applied at perturbation level only: `adotoa_matter` uses $H_{\text{matter}}^2(a) = H_{\Lambda\text{CDM}}^2 + \beta_m \mathcal{E}(a) H_0^2$. Global background (`dtauda`) unchanged.

#	Test	Result	Key Diagnostic
1	Planck best-fit $\Delta\chi^2$	PASS	+0.54 vs. ΛCDM
2	σ_8 suppression	PASS	0.8087 $\rightarrow$ 0.7998; direction correct
3	Standard-param stability	PASS	All shifts $< 0.1\sigma$
4	β_m posterior	PASS	0.157 ± 0.029 ; 0.3% from virial
5	$R - 1$ convergence	PASS	< 0.01 all 3 chains
6	Background-modification diagnostic	PASS	$H_0 \approx 61.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ confirms perturbation-only mechanism
7	Sector split confirmed	PASS	$\beta_\gamma/\beta_m < 8.5 \times 10^{-6}$

4 Section 3: Deterministic Validation Script (`iam_validation.py`)

Dataset: 10 data points (3 H_0 + 7 DESI growth rates). **Runtime:** ~ 1 min. **Status:** **CONFIRMED**.

#	Test	Result	Key Diagnostic
1	ΛCDM baseline χ^2	PASS	$\chi_{H_0}^2 = 31.91$; $\chi_{\text{growth}}^2 = 6.36$; total = 38.28
2	IAM dual-sector χ^2	PASS	$\chi_{H_0}^2 = 1.52$; $\chi_{\text{growth}}^2 = 6.75$; total = 8.27
3	Improvement	PASS	$\Delta\chi^2 = 30.01$ (5.5σ); $\Delta\chi_{H_0}^2 = 30.40$
4	Profile likelihood	PASS	$\beta_m = 0.157 \pm 0.029$ (68%); ± 0.058 (95%)
5	Photon-sector constraint	PASS	$\beta_\gamma < 0.004$ (95% CL)
6	CMB lensing consistency	PASS	Growth suppression 1.36%; compensates $\sim 85\%$ of θ_s shift
7	Model selection	PASS	$\Delta\text{AIC} = 26.01$; $\Delta\text{BIC} = 25.40$; $\mathcal{L}(\Lambda\text{CDM})/\mathcal{L}(\text{IAM}) = 2.25 \times 10^{-6}$
8	Full MCMC analysis	PASS	$\beta_m = 0.157_{-0.029}^{+0.029}$; $H_0(\text{matter}) = 72.5 \pm 1.0$
9	Pantheon+ SNe consistency	PASS	Primary IAM impact on growth, not geometry

5 Section 4: Derivation Verification Suite (iam_derivation_tests.py)

Runtime: 0.1 sec. **All 15 tests PASS.** Executed March 5, 2026.

Physics note (corrected March 5, 2026): $H_{\text{matter}}^2(a)$ is the matter-sector perturbation friction only. The background Friedmann equation remains standard Λ CDM throughout. All μ -ratios are $E_{\Lambda\text{CDM}}^2/H_{\text{matter}}^2$, not a modified background.

Derivation Chain (Tests 1–10)

#	Test	Result	Key Diagnostic
1	Jacobson (1995): standard entropy $\rightarrow$ Friedmann	PASS	$E^2(a=1) = 1.0000000000$; Raychaudhuri match True
2	Cai–Kim (2005): first law on apparent horizon	PASS	Flatness = 1.0000000000; $T_H/H = 0.159155$
3	Modified entropy $\rightarrow$ matter-sector friction	PASS	$H_{\text{matter}}^2(1) = 1.157500 = 1 + \beta_m$; $H_0(\text{matter}) = 72.51 \text{ km s}^{-1} \text{ Mpc}^{-1}$; $E_{\Lambda\text{CDM}}^2(1) = 1.000$ (unchanged)
4	Cumulative decoherence integral $\rightarrow e^{\alpha-\beta/a}$	PASS	Best $n=3.5$: $\alpha = 2.283$, $\beta = 1.009$, $r = 0.644$; β crosses 1.0 in $n \in [3, 4]$
5	Sheth–Tormen collapse rate $\rightarrow 1/a$ coefficient	PASS	Best $\sigma^*=1.5$: $\beta = 1.058$ (5.8% from unity, galaxy-group scale)
6	Virial theorem $\rightarrow$ $\beta_m = \Omega_m/2$	PASS	$\beta_m = 0.1575$ (predicted) vs. 0.157 (MCMC): 0.3% agreement
7	Collapsed fraction confirms virial	PASS	$f_{\text{coll}} = 0.5927$; naive $\beta = 0.1867$ (19% off); virial $\beta = 0.1575$ (0.3% off)
8	Perturbation theory: $\mu < 1$, $\Sigma = 1$	PASS	$\mu(z=0) = 0.8639$; $\mu(z=0.5) = 0.9482$; $\mu(z=1) = 0.9823$; $\mu(z=10) = 1.000$; $\Sigma = 1$ exact
9	Fixed $\beta_m = \Omega_m/2$: zero parameters	PASS	$\chi_{\Lambda\text{CDM}}^2 = 41.43$ vs. $\chi_{\text{IAM}}^2 = 10.20$; $\Delta\chi^2 = 31.2$ (5.6σ); ΛCDM $6 \times 10^6 \times$ less likely; $H_0(\text{matter}) = 72.51$ (0.51σ from SH0ES)
10	Equation of state: $w_{\text{info}} = -1 - 1/(3a)$	PASS	$w_{\text{eff}}(z=0) = -1.0623$ (phantom); mapped $w_0 = -1.070$, $w_a = +0.042$; DESI confirms DE evolution

Robustness Checks (Tests 11–15)

#	Test	Result	Key Diagnostic
11	Continuity equation: $\dot{\rho} + 3H(1+w)\rho = 0$	PASS	Max residual 7.60×10^{-16} (machine precision); identity holds for all $a \in [0.1, 2.0]$
12	MGCAMB pure_MG parametrization accuracy	PASS	Max error 4.29 pp at $z=0$; adequate for $\sigma(\mu_0) \sim 0.2$ current sensitivity
13	Sensitivity: μ_0 vs. exponent $c \in [0.8, 1.2]$	PASS	μ_0 spread = 0.00000 (insensitive to exponent); σ_8 spread 0.0031 (0.4%)
14	Sensitivity: μ_0 vs. coupling β_m	PASS	$\beta_m = 0.1575$ sits at minimum-tension point (0.51σ from SH0ES); virial prediction optimal
15	Reparametrization: IAM $\neq w_0 w_a$ CDM	PASS	$H(z)$ degenerate to 7.05%; $\mu(z)$ completely different (IAM $\mu < 1$, $w_0 w_a \mu = 1$); growth breaks degeneracy

Derivation chain (Tests 1–10): Jacobson (1995) $\rightarrow$ Cai–Kim (2005) $\rightarrow$ Modified entropy (S_{info}) $\rightarrow$ Cumulative integral $\rightarrow$ Sheth–Tormen $\rightarrow \beta_m = \Omega_m/2 \rightarrow \delta\varphi = 0 \rightarrow$ perturbation friction $H_{\text{matter}}^2 \rightarrow \mu < 1, \Sigma = 1$ (background = Λ CDM). $\Delta\chi^2 = 31.2$ (5.6σ , 0 free parameters).

6 Section 5: Empirical Evidence Inventory

#	Observable	Status	Notes
1	H_0 tension	CONFIRMED	$H_0(\text{matter}) = 72.51 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (0.51 σ)
2	σ_8 tension	CONFIRMED	$\sigma_8 = 0.87 \rightarrow 0.7998$ (perturbation mechanism)
3	β_m posterior	CONFIRMED	0.157 ± 0.029 (virial: 0.1575; 0.3% match)
4	Sector split	CONFIRMED	$\mu_{\text{eff}}/\beta_m < 8.5 \times 10^{-6}$
5	$\Sigma = 1$ (photon sector)	CONFIRMED	$\delta\varphi = 0$; BAO positions identical to ΛCDM
6	DESI growth rates	CONSISTENT	Not suppression direction correct
7	Planck CMB TT	CONSISTENT	$\Delta\chi^2 < 3.84$ all Level 1 combos; +0.54 Level 2a
8	Pantheon+ SNe	CONSISTENT	Distance-geometry unchanged (photon sector)
9	CMB lensing	CONSISTENT	Growth suppression compensates $\sim 85\%$ of θ_s shift
10	Cluster $M_{\text{lens}}/M_{\text{dyn}}$	PENDING	Predicted = $1/\mu(z)$; Euclid/eROSITA
11	Euclid DR1 μ_0	PENDING	$\sim 3.4\sigma$ projected; ~ 8 months from early 2026
12	DESI Year 5	PENDING	Growth rate falsification
13	CMB-S4 lensing	PENDING	ISW cross-correlation; $\Phi + \Psi$
14	ISW cross-correlation	PENDING	Reduced ISW amplitude predicted
15	E_G statistic	PENDING	RSD-lensing cross; IAM predicts $E_G < E_{G\Lambda\text{CDM}}$
16	Bulk flows	PENDING	Suppressed at $> 100h^{-1} \text{ Mpc}$
17	Void statistics	PENDING	VOIDS larger/emptier under $\mu < 1$
18	BAO peak positions	CONFIRMED	Photon-sector observable; identical to ΛCDM

Summary: 5 confirmed / 3 consistent (non-unique) / 4 pending near-term tests / 4 pending long-baseline tests.

7 Section 6: Paper Pipeline Status

Paper	Status	Notes
Observational constraints (main paper)	Submitted	Universe journal ID universe-4189350; desk-rejected without review
IAM Theory Paper	Complete	First-principles derivation chain
Koide relation (conditional theorem)	In revision	7 referee items; Oxford outreach pending
Master Paper (IAM_Master_Paper.tex)	Complete	~1,721 lines; submission-ready
$w(z)$ / Far Future paper	Pending	Rigorous/testable
Fruit Theory paper	Pending	Speculative
Walking With Giants	Done	Verify outputs

8 Section 7: arXiv Endorsement and Outreach Status

Contact / Venue	Status	Notes
Zhuangfei Wang (MGCAMB)	No response	Batch 1, sent Feb 13, 2026
Levon Pogossian (μ - Σ)	No response	Batch 1, sent Feb 13, 2026
Alessandra Silvestri (μ - Σ)	No response	Batch 1, sent Feb 13, 2026
Euclid/ μ - Σ contacts (9)	Not yet sent	Batch 2 emails pending
Koide contacts (Nishiura, Koide, Rivero)	Pending	After Koide revision
Rodejohann, Foot	High priority	arXiv endorsement path
arXiv endorsement	Pending	Primary bottleneck
Universe journal	Desk-rejected	Without reasons

9 Section 8: Derivation Suite Output (Exact Script Output)

The following is the exact terminal output of `iam_derivation_tests.py` run March 5, 2026, 15/15 PASS, runtime 0.1 seconds. All numbers reproduced verbatim from the printout.

Derivation chain summary:

- Test 1: $E^2(a=1) = 1.0000000000$; Raychaudhuri match **True**
- Test 2: Flatness = 1.0000000000; $T_H/H = 1/(2\pi) = 0.159155$
- Test 3: $H_{\text{matter}}^2(a=1) = 1.157500 = 1 + \beta_m$; $H_0(\text{matter}) = 72.51 \text{ km s}^{-1} \text{ Mpc}^{-1}$; $E_{\Lambda\text{CDM}}^2(a=1) = 1.0000000000$ (unchanged)
- Test 4: Best $n = 3.5$: $\alpha = 2.283$, $\beta = 1.009$, $r = 0.643533$; both cross unity in $n \in [3, 4]$

- Test 5: Best $\sigma^* = 1.5$: $\beta = 1.058$ (5.8% from unity)
- Test 6: $\beta_m = \Omega_m/2 = 0.315/2 = 0.1575$; MCMC 0.157 ± 0.02 ; agreement 0.3%
- Test 7: $f_{\text{coll}} = 0.5927$; naive $\beta = 0.1867$ (18.9% off); virial $\beta = 0.1575$ (0.3% off)
- Test 8: $\mu(z=0) = 0.8639$; $\mu(z=0.5) = 0.9482$; $\mu(z=1) = 0.9823$; $\mu(z=10) = 1.000$; $\Sigma = 1.0$ exactly
- Test 9: $\chi^2(\Lambda\text{CDM}) = 41.43$; $\chi^2(\text{IAM}) = 10.20$; $\Delta\chi^2 = 31.2$ (5.6σ); LCDM $6,038,848\times$ less likely; $H_0(\text{matter}) = 72.51 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (0.51σ from SH0ES)
- Test 10: $w_{\text{eff}}(z=0) = -1.0623$; $w_{\text{eff}}(z=1) = -1.0520$; mapped $w_0 = -1.070$, $w_a = +0.042$
- Test 11: Max fractional residual 7.60×10^{-16} (machine precision)
- Test 12: Max error 4.29 pp at $z=0$; adequate for $\sigma(\mu_0) \sim 0.2$
- Test 13: μ_0 spread = 0.00000 over $c \in [0.8, 1.2]$; σ_8 spread 0.0031 (0.4%)
- Test 14: $\beta_m = 0.1575$ gives H_0 tension 0.51σ (minimum-tension point); $\beta < 0.14$: insufficient; $\beta > 0.18$: overcorrects
- Test 15: $H(z)$ degenerate to 7.05%; $\mu_{\text{IAM}}(z=0) = 0.8639$ vs. $\mu_{w_0w_a} = 1.0000$; growth breaks degeneracy

Reproducibility

All validation code, chain files, YAML configs, and data: <https://github.com/hmahaffeyges/IAM-Validation>

Last updated: March 5, 2026